

Hannah Hsiao

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EDUCATION

Georgia Institute of Technology, Atlanta, GA

expected May 2027

Master of Science in Computer Science, Concentration in Artificial Intelligence

GPA: 4.0

Bachelor of Science in Computer Science, Concentration in Intelligence and Devices

Aug 2023 — May 2026

- Relevant Coursework: Data Structures and Algorithms, Linear Algebra, OOP, Computer Organization, Systems and Networks, Artificial Intelligence, Machine Learning, Natural Language Processing, Computer Vision

EXPERIENCE

Software Engineering Intern

Austin, TX

IBM, Electronic Design Automation (EDA)

May 2026 – August 2026

- Engineered an AI-assisted debugging workflow and MCP-based self-service tools that reduced environment troubleshooting cycles by 98%, enabling engineers to diagnose configuration issues without asynchronous support
- Optimized an infrastructure routing algorithm that automated a previously manual server configuration workflow using server utilization and queue depth to dynamically redirect workloads and prevent system overload

Graduate Teaching Assistant

Atlanta, GA

CS 3510: Design & Analysis of Algorithms, Georgia Institute of Technology

August 2025 — present

- Provide algorithm design and problem-solving support to 600+ students through office hours, covering graphs, dynamic programming, divide and conquer, and complexity analysis; develop original algorithm design problems

Software Engineering Intern

Dearborn, MI

Ford Motor Company, Fleet Management Team

May 2025 – August 2025

- Spearheaded adoption of **Playwright** for end-to-end testing across three fleet management teams, integrating automated tests into the CI/CD pipeline and improving release efficiency by 40% while maintaining reliability
- Developed a scalable **React** front-end enabling fleet managers to request vehicle titles, renew registrations, and track progress in real time, reducing customer task completion time by 80%
- Integrated **REST APIs** into a **Spring Boot** backend to enable secure data synchronization and automated title request tracking, improving workflow transparency and reducing manual follow-ups

Undergraduate Teaching Assistant, Forum Lead

Atlanta, GA

CS 1331: Object-Oriented Programming in Java, Georgia Institute of Technology

January 2024 — May 2025

- Led weekly recitations for 30 students and managed technical support for a course of 800+ students, explaining Java and object-oriented programming concepts including inheritance, polymorphism, recursion, and software design

Undergraduate Researcher

Atlanta, GA

RAIL Lab, Georgia Institute of Technology

August 2024 — May 2026

- Designed and executed experiments to benchmark autonomous robotic grasping pipelines on the Hello Robot Stretch RE 2 platform, comparing geometry-driven computer vision heuristics with end-to-end imitation-learning models
- Developed a **ROS**-based evaluation pipeline to collect precise gripper accuracy measurements across manipulation trials and quantitatively compare model performance
- Analyzed algorithmic and sensor-level failure modes in point cloud segmentation and depth estimation, identifying limitations caused by partial observability and surface specularities; documented findings in an undergraduate thesis

PROJECTS

Doc McQuery | Python, PostgreSQL, pgvector, Sentence Transformers, REST API, OpenAI API

Georgia Tech HackGT, 2nd place in Healthcare Track

- Competed against over 200 teams to build a **RAG-based semantic search system** for medical case reports using **Sentence Transformers** to generate embeddings and **pgvector** for vector similarity retrieval
- Designed retrieval and prompt workflows that transformed natural language queries into relevant document context and generated context-aware medical summaries using the **OpenAI API**
- Built a full-stack **React** and **Flask** web application with interactive search results and side-by-side case comparison

Breadboard Analyzer | Python, OpenCV, React, Flask

- Developed a web application using **React** that converts uploaded breadboard images into corresponding logic circuit schematics using classical computer vision techniques
- Implemented an **OpenCV** image-processing pipeline using grayscale conversion, edge detection, and contour analysis to identify wires and integrated circuits for schematic generation

SKILLS AND COMPETENCIES

Tools: Git, Linux, Docker, OpenCV, CMake, Flask API, CI/CD, ROS, PostgreSQL, Unity, Figma

Languages/Frameworks: Python, C/C++, Java, JavaScript, PyTorch, NumPy, React.js, Spring Boot, HTML, CSS, C#